

COSC2148 / COSC3130 — Computing Research & Project Preparation

Assignment 2 — Draft Project Proposal

Using Physiological Cues to Improve Empathy and Human-AI Interaction in a Mixed Reality Environment

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1 Introduction

Emotion is a fundamental dimension of human experience — a bounded, multicomponent phenomenon in which physiological changes, facial expressions, and bodily responses synchronise in response to events that are relevant to an individual’s needs and goals [1]. As human-computer interaction (HCI) becomes increasingly embedded in daily life, the capacity of computational systems to recognise and respond to these emotional states has emerged as a critical frontier [2]. The field of affective computing pursues this goal: developing systems that can detect human affect from observable signals and adapt their behaviour accordingly [2]. Physiological signals — including electrodermal activity (EDA, also referred to as galvanic skin response), heart rate (HR), facial expressions, and eye gaze — have been established as reliable indicators of emotional states, each capturing distinct dimensions of the autonomic and somatic responses that accompany emotion [3–6].

The deployment of affective computing systems in immersive environments has accelerated alongside advances in Virtual Reality (VR), Augmented Reality (AR), and extended reality (XR) technology. Since Milgram and Kishino [7] first defined the reality-virtuality continuum, Mixed Reality (MR) — and the broader family of VR, AR, and XR — has evolved from a theoretical taxonomy into an active research platform. Systematic reviews of empathic computing in immersive contexts confirm that physiological sensing [8] and responses driven by Artificial Intelligence (AI) [9] have both been demonstrated in VR settings. However, MR — which blends real and virtual elements in a shared environment — remains substantially underexplored as a context for empathic interaction between human and AI [9, 10].

A critical limitation in current affective computing systems is their reliance on generalised physiological models. Across VR and wearable contexts, user-independent models consistently underperform personalised alternatives, with accuracy gaps of up to 30 percentage points reported in direct comparisons [11, 12]. This reflects individual response specificity [13]: the same emotion produces different physiological patterns in different people, making population-average models unreliable. While personalised classifiers have demonstrated clear accuracy advantages [14, 15], their seamless integration with real-time empathetic agent responses has not been achieved. Existing empathic agent systems in conversational and VR contexts demonstrate that physiological signals can drive more empathic interactions [16, 17], yet even the most advanced implementations acknowledge that physiological classification is performed once rather than continuously throughout the interaction [16, 18].

This project investigates the combination that the literature has not yet addressed: a personalised, real-time physiological classification system connected to an empathic AI agent in a Mixed Reality environment. Three research questions guide the investigation, addressing the reliability of real-time physiological detection, the effect of continuous empathic response on user experience, and the relationship between physiological and subjective measures of empathy.

2 Research Questions

This project addresses three interconnected research questions that, together, target the gap identified in the literature: the absence of a system that combines personalised, real-time physiological classification with continuous empathetic agent response in a Mixed Reality environment. The questions align with a detect $\rightarrow$ respond $\rightarrow$ evaluate pipeline, with each question investigating one stage of the mechanism.

RQ1 (Detection): How reliably can real-time multimodal physiological classification, personalised through a brief pre-session calibration, transfer to infer users’ affective states during immersive Mixed Reality (MR) interaction?

RQ1 targets the foundational detection layer. Extensive research has been conducted using physiological signals to detect emotional states, and multiple studies report personalised classifiers achieving substantially higher accuracy than user-independent alternatives trained on the same data, with accuracy gaps typically in the range of 15 to 30 percentage points across VR and wearable contexts [11, 12, 14, 15]. Chioffi et al. [19] further demonstrated that universal electrodermal activity thresholds failed for approximately 20% of participants in a closed-loop VR study, reinforcing that individual calibration is a design requirement rather than an enhancement [13]. However, these results have been established in seated VR or desktop contexts where calibration and classification occur in the same environment. RQ1 specifically tests whether calibration data collected pre-session transfers to affective classification during embodied MR interaction — an open empirical question for any system where calibration and deployment conditions differ [9, 20].

RQ2 (Response): Does an empathic AI agent that continuously adapts its verbal responses to users’ real-time physiological states produce a measurably different interaction experience compared with a non-adaptive baseline in Mixed Reality?

RQ2 extends the detection layer into a closed interaction loop, in which physiological signals are continuously sampled, classified into affective states, used to adapt system behaviour, and then re-sampled as the user responds to that adaptation. Several studies report that such closed-loop physiological adaptation produces measurable user outcomes in immersive settings, including improved working memory, sustained engagement, and higher subjective ratings of the agent [17, 19, 21]. Most directly, Chang et al. [10] reported significant increases in perceived empathy, trust, and social connectedness when an MR agent acknowledged user skin conductance changes compared to a non-adaptive baseline — an agent that produced scripted responses independent of user state. However, no existing system has combined continuous real-time classification with generative empathic response in MR: Chang varied agent awareness experimentally rather than driving it from a live classifier, and Dongre et al. [18] explicitly identify continuous reclassification as the unmet next step. Recent work on Large Language Model (LLM) affective

conditioning through prompt injection [22, 23] suggests generative response adaptation is tractable; RQ2 tests whether that tractability translates into a measurable experiential difference when embedded in MR.

RQ3 (Evaluation): To what extent do real-time physiological measures of affect align with users' subjective experience of empathy, trust, and presence during empathic human-AI interaction in Mixed Reality, and what does this imply for evaluation methodology in this domain?

RQ3 addresses a measurement validity gap that prior work has identified but not resolved. Prior empathic agent studies typically report either physiological outcomes or subjective questionnaires, rarely integrating both in the same analysis, and none have conducted such integration in MR with a personalised classifier as the mediating mechanism [10, 16, 17, 24]. This matters methodologically: if physiological signals and subjective experience diverge, affective systems may be optimising for signals that do not reflect user experience; if they converge, physiological measures are validated as an efficient proxy for subjective evaluation in MR. This investigation will triangulate objective physiological data against the Perceived Empathy of Technology Scale (PETS) [25], the Almere acceptance model [26], and the Igroup Presence Questionnaire (IPQ) [27], the latter following modern scoring guidance [28] to ensure comparability with prior MR work.

3 Literature Review

The literature review will draw on prior research to identify the gap this project addresses, tracing four threads: the maturity of physiological emotion sensing; the emergence of VR and Mixed Reality as active research areas conveying affective information that is difficult to voluntarily suppress; and the remaining integration gap.

Physiological Signals as Emotion Indicators. The scientific basis for using physiological signals as indicators of emotion is well established. Early evidence from Ekman et al. [4] demonstrated that distinct patterns of facial muscle activity reliably correspond to discrete emotional states, indicating that emotion manifests in observable, measurable form. Subsequent work extended this insight to autonomic and central nervous system signals. Electrodermal activity (EDA) and heart rate (HR) track emotional arousal continuously through the autonomic nervous system [3], while electroencephalography (EEG) captures cortical correlates of affective processing with comparable temporal fidelity [29]. Eye gaze has more recently emerged as a complementary modality, with fixation duration, pupil diameter, and scan path patterns carrying affective information that is difficult to suppress voluntarily [5]. A comprehensive systematic review of 386 studies using peripheral physiological signals confirms that these modalities now dominate the emotion recognition literature, with multimodal combinations consistently outperforming single-signal methods [6]. Marín-Morales et al. [24] demonstrated that this multimodal approach extends to immersive VR environments using portable sensors capturing brain and cardiac activity. Taken together, these findings establish a foundation for detecting emotion through physiological signals and for VR as an ecologically valid environment for capturing them.

The VR/XR Research Context. The deployment of affective sensing in immersive environments has grown substantially over the past decade. Reviews of VR for emotion elicitation confirm that immersive environments reliably induce measurable affective states, as evidenced by stable, interpretable physiological signals [8]. The intersection of XR and AI has been characterised as a rapidly expanding research frontier, with affective adaptation identified as a promising application [30]. MR specifically has gained traction as consumer-grade mixed-reality headsets have become widely available. Yet despite this momentum, systematic reviews consistently identify MR as underexplored relative to VR for empathic applications: a review of 77 papers on empathic computing in immersive environments from 2000 to 2024 found VR dominating the literature [9], and a complementary systematic review of 23 embodied conversational agent studies in XR found that physiological sensing is rarely integrated — only one of the 23 papers used heart rate, and none used multimodal physiological classification [20]. This distinction matters for the sensing problem. VR studies typically keep participants seated in a controlled setup. MR users, by contrast, interact physically with real spaces while engaging with virtual agents, which imposes additional constraints on the quality of physiological signals.

The Generalisation Problem. One significant technical obstacle in affective computing for immersive environments is individual variability. Fairclough [13] identified individual response specificity as a fundamental design challenge: the same emotion produces different physiological patterns in different people, meaning that models trained on population-average data will consistently misclassify individuals. This theoretical claim has since been empirically confirmed across multiple contexts. In VR, Lima et al. [11] found that user-dependent models achieved 69–71% accuracy on five-class emotion classification, while user-independent models using the same signals degraded to 54% and, in some conditions, to 24.9%. On consumer wearables, Li and Washington [12] reported a similar pattern: personalised models reached 95% accuracy, compared to 67% for generalised equivalents. Fu et al. [15] attribute this gap to distributional shifts between individuals’ physiological signals and propose transfer learning as a mechanism for adaptation. The systematic review by Gkintoni et al. [29] identifies cross-subject generalisation as one of the top unsolved challenges in the field, noting a consistent lab-to-real-world performance degradation that is compounded by individual variability. This indicates that accurate physiological emotion classification for individual users requires personalised models rather than generalised ones. Chiossi et al. [19] implemented a closed-loop physiological adaptation system in VR and found that universal electrodermal activity (EDA) thresholds failed for approximately 20% of participants. This reinforces the argument that individual calibration is not optional.

From Closed Loop to Empathic Agent: The Remaining Gap. Although personalised classification is now well-established, its connection to real-time empathic agent response has not been achieved in the MR context. In adjacent domains, the literature shows that systems can effectively adapt their behaviour by continuously sampling physiological signals, classifying them, and adjusting in response as the user’s state evolves. Ewing et al. [21] demonstrated this in gaming, using real-time EEG to modulate game difficulty and sustain player engagement. Chiossi et al. [19], the closest existing precedent for the proposed system, achieved the same in a social VR environment, where EDA-driven adjustment of environmental complexity improved working memory performance.

However, neither study extended this adaptation to an empathic agent that would continuously use physiological signals to adjust its verbal response. Saffaryazdi et al. [16] developed a conversational agent that detects emotion from neural and physiological signals in real time and generates empathic responses, reporting higher empathy and engagement scores in the physiologically-aware condition. The authors note, however, that their agent did not use the recognition result to continuously adapt its behaviour. Dongre et al. [18] acknowledge the same limitation in their Large Language Model (LLM)-based EmLLM framework, identifying continuous reclassification as the critical next step not yet achieved. A parallel question is whether LLMs can adapt generative responses to affect: Schmidmaier et al. [22] showed in a user study of 200 participants that multi-modal LLMs conditioned on affective context produce linguistically distinct responses, while Fazzi et al. [23] documented the limits of this approach,

finding that LLMs struggle to maintain high-intensity emotional states over extended dialogue. Together, these show that LLM-based generative empathetic responses are viable with careful prompt design.

In MR specifically, Chang et al. [10] provide the most direct precedent, demonstrating that agent awareness of a user’s physiological state significantly improves perceived empathy, social connectedness, and trust. Yet this awareness was varied experimentally rather than driven by a real-time personalised classifier. AffectivelyVR [14] and CAEVR [17] independently demonstrate each half of the problem in VR — personalised physiological classification in one case, empathic agent response in the other — but no system has combined them continuously in Mixed Reality.

The proposed project addresses this gap. It combines a system that recognises emotion from each user’s physiological signals with an AI agent that uses this signal continuously to produce empathic responses in a Mixed Reality environment. The outcome will be evaluated using both objective physiological signals and established questionnaires covering empathy [25], trust and acceptance [26], and presence [27, 28]. This will provide the first empirical test of whether adapting AI responses to each user’s physiological signals produces measurable improvements in interaction quality in MR.

4 Research Methodology

This project will follow a design-based research (DBR) methodology [31], in which iterative cycles of design, implementation, analysis, and reflection are used to refine an innovative system. Closely related work in biocybernetic loop development [21] and empathic XR systems [18] has applied the same iterative pattern.

Addressing RQ1 (Detection). RQ1 asks how reliably a personalised multimodal classifier can transfer to infer affective states during MR interaction. Prior work has addressed similar detection questions by combining data from multiple physiological sensors and calibrating them to individual participants using pre-rated affective stimuli [14, 17]. This project will capture electroencephalography (EEG), electrodermal activity (EDA), and eye gaze throughout each session, beginning with a brief per-participant calibration using validated affective stimuli. Each session will also include a short baseline acquisition inside the MR environment without any affective content, providing a per-participant reference point for normalising classifier output during the main interaction. Classification accuracy on held-out calibration data will provide the primary answer to RQ1.

Addressing RQ2 (Response). RQ2 asks whether a physiologically-adaptive empathic agent produces a different interaction experience from a non-adaptive baseline in MR. Prior work has addressed similar questions by comparing agents that respond to physiological states against non-adaptive baselines matched for conversational surface [10, 16]. Building on these precedents, this project will deploy two agent conditions, each experienced by every participant: an LLM-based adaptive agent [18, 22], and a non-adaptive baseline independent of user physiological state [10]. The task context is a tense exploratory MR scenario with horror elements, a paradigm shown to produce measurable physiological stress responses [32–34]. Comparison of subjective experience measures between the two conditions will provide the primary answer to RQ2.

Addressing RQ3 (Evaluation). RQ3 asks how closely real-time physiological measures align with users’ subjective experience during empathic human-AI interaction in MR. Prior empathic agent studies typically report physiological outcomes or subjective questionnaires in isolation [16, 17, 24], with no integrated analysis in MR [10]. Building on these precedents, this project will analyse, in an integrated form, data streams from the same participants, combining real-time classifier output with post-session responses covering empathy [25], trust and acceptance [26], and presence [27, 28]. Convergence would validate physiological signals as efficient evaluation proxies, while divergence would argue for multi-measure evaluation in future MR empathy research.

5 Evaluation

Scope. This project investigates whether personalised multimodal physiological classification can drive an empathic AI agent’s responses in MR, producing measurable differences in interaction experience compared to a non-adaptive baseline.

Deliverables. An operational prototype integrating physiological sensing with an adaptive LLM-based agent in an MR environment; main study results include classifier performance and subjective interaction measures documented as a thesis. Commercial deployment and modalities outside the EEG, EDA, and eye-gaze signal set are out of scope.

Timeline and milestones. Table 1 summarises the work breakdown into major phases, and Figure 1 shows the corresponding timeline across the 28-week two-semester project window. Phase-transition checkpoints occur at weeks 4, 12, 16, and 24, each requiring sign-off from the project lead before the next phase begins. The critical path runs through design finalisation → ethics approval → technical pilot → main data collection, as delays in any of these stages block all subsequent work. Weekly team reviews run throughout the project, covering progress, risk status, and upcoming transitions.

Phase	Key activities
Design and preparation	Finalise experimental design; MR scenario design; recruitment materials
Ethics and build	Ethics submission and review (parallel); build MR environment, agent pipeline, and sensing/classifier
Validation	Technical pilot (sensing pipeline); process pilot (interaction flow)
Data collection and analysis	Main study across participants; classifier and subjective analysis
Write-up	Thesis documentation

Table 1: Work breakdown by phase.

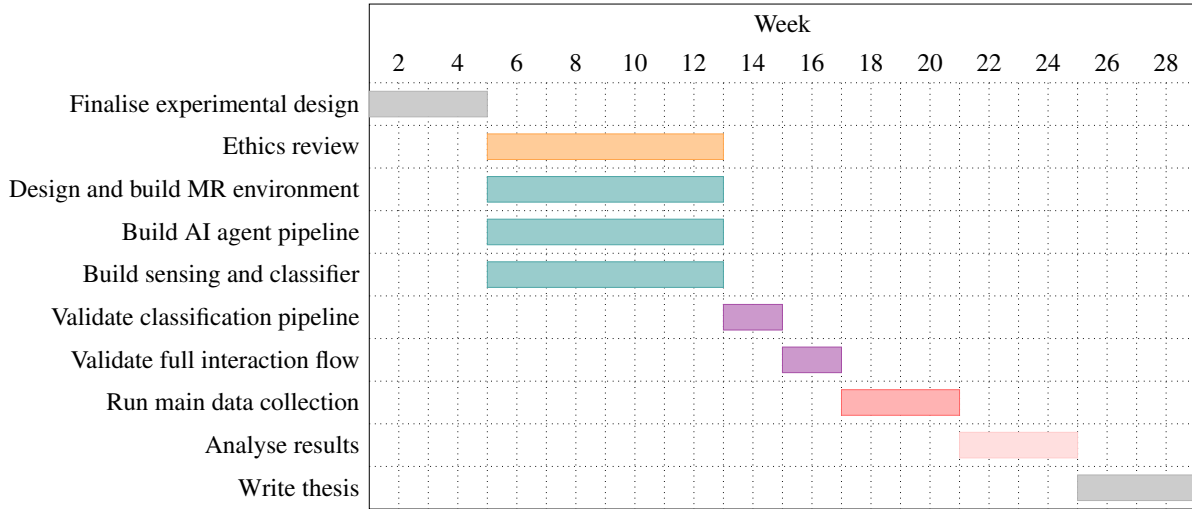


Figure 1: Research execution timeline across the 28-week project window.

Task ownership.

Task area	Owner	RQ linkage
Physiological sensing and classifier pipeline	Vishal	RQ1
AI agent and LLM conditioning	Jaswanth	RQ2
Human interaction testing and user study protocol	Anthony	RQ2
MR scenario design and team coordination	Kennosuke (lead)	RQ2
Data analysis and risk management	Yashas	RQ3
Final report compilation	Team	All

Table 2: Task ownership and linkage to research questions.

Risk management. Table 3 presents the identified risks using a cause → event → impact structure, with probability, impact, priority, response strategy, mitigation actions, and the generic role responsible for each risk.

ID	Risk description (cause → event → impact)	Probability	Impact	Priority	Response strategy	Mitigation actions	Owner
R1	Ethics committee queue time and review complexity → ethics approval delayed beyond week 12 → pilots and main study cannot commence	Mid	High	Critical	Mitigate	Prepare submission during weeks 1–4; submit week 5; keep build workstreams independent so they continue during review	Project Lead
R2	Three independent build streams (sensing, agent, MR environment) → components fail to integrate into a working end-to-end system → main study cannot run	Mid	High	Critical	Mitigate; Avoid	Technical pilot (weeks 13–14) isolates integration issues before main study; weekly integration checks during build; modular architecture so components can be demonstrated independently if full integration fails	Technical Lead
R3	Domain mismatch (desktop calibration → embodied MR deployment) and individual variability → classifier performance below usable threshold → RQ1 detection fails and agent cannot be driven by meaningful signals	Low	High	High	Mitigate; Avoid	Two-stage calibration mitigates domain mismatch; precedent studies demonstrate feasibility; benchmark comparison contextualises performance; fallback to simpler physiological-state adaptation if classifier underperforms	Classifier/Sensing Lead
R4	Sensitive physiological data with insufficient access controls → unauthorised access or data loss → ethics breach and participant harm	Low	High	High	Mitigate; Avoid	Follow institutional ethics protocols; anonymisation at collection; encrypted storage; access restricted to research team; collect only data methodologically required	Data Manager
R5	Horror-scenario intensity and session length → participants withdraw before completing both conditions → reduced sample size and potential statistical underpower	Low	Mid	Mid	Mitigate; Avoid	Recruit 25 to target 20–24 completions; clear participation agreement; calibrate scenario to mild-stress range; pre-screen participants; allow pause or withdrawal without penalty	User Testing Lead

Table 3: Key project risks with cause, event, impact, response strategy, mitigations, and owner role.

For each risk: **before occurrence**, prevention through the listed mitigations combined with weekly integration checks during implementation phases; **when occurring**, immediate escalation per the path below and activation of contingency workstreams (extended ethics review window, fallback sensor configurations, reduced-N analysis with documented limitations); **after occurrence**, documented lessons incorporated into subsequent phases and handover documentation.

Escalation path: task owner → project lead (Kennosuke) → supervisor. Risk status reviewed weekly.

Methods and tools. The project uses a Kanban-style approach to day-to-day task management. This approach suits research work where task durations are heterogeneous, and some dependencies (e.g., ethics approval) are beyond the team’s control. Scheduling is anchored by the Gantt in Figure 1 and phase checkpoints. Progress will be tracked on Trello, used as a shared Kanban board with owner assignments and status columns, and communication will run through Slack for day-to-day coordination and weekly team reviews for synthesis. Reporting and version control use a shared GitHub repository accessible to all team members.

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Appendix: Generative AI Declaration

This work was primarily human-created. AI was used to make stylistic edits, such as changes to structure, wording, and clarity. AI was used to make content edits, such as changes to scope, information, and ideas. AI was prompted for its contributions, and AI assistance was enabled. AI-generated content was reviewed and approved. The following model(s) or application(s) were used: **Claude (Anthropic)** and **Grammarly**. AIA AI v1.0

Detailed per-tool documentation follows in Tables 4 and 5 below.

Table 4: Generative AI Tool Usage — Claude

Field	Description
1. GenAI tool used	Claude (Anthropic), accessed via the Claude web interface.
2. Purpose of GenAI tool for this assignment	- Brainstorming research questions, methodology options, and section outlines - Drafting content for revised sections - Rewriting, editing, and checking alignment against the brief across multiple review passes
3. Prompts or instructions given	Representative examples: “Can you go through the draft and suggest any sections that require revision?” “Can you help me tighten up this paragraph?” “We have to change the wording for p1 because it overuses the word loop”
4. Amount of prompts or instructions given	Approximately 50+ prompts across multiple sessions over several days.
5. Summary of GenAI output	Claude proposed structural decisions, refined draft sentences, suggested the organisation of the by-RQ methodology, generated LaTeX formatting, including the pgfgantt timeline, and performed brief alignment audits, identifying gaps against rubric requirements. Example excerpt: “This project investigates the combination that the literature has not yet addressed: a personalised, real-time physiological classification system connected to an empathic AI agent in a Mixed Reality environment.”
6. GenAI output verification	All content was verified against source materials by the research team. Every cited paper was present in the user’s Zotero collection and had been read. Specific numeric claims (accuracy percentages, study sample sizes, paper counts) were cross-checked against source PDFs. Methodological and structural choices were approved by the user based on their own reading.
7. GenAI output usage or modification	The user reviewed every paragraph, frequently rewriting sentences for clarity or register, cutting jargon-dense phrasing, pushing back on methodological suggestions that overclaimed, and restructuring sections flagged for logical inconsistencies. Multiple review passes were conducted with substantive rewording applied in each.
8. Reflection on generated content and usage	What worked well: Claude was effective in refining the initial drafts to build the foundation of the content. It was particularly useful for identifying logical inconsistencies between sections, catching brief-alignment gaps, and tightening wordy phrasings. What did not work well: Claude occasionally overclaimed about source papers and introduced invented terminology when trying to avoid problematic phrases, requiring the user to cross-check and correct. Value added: The collaborative drafting approach significantly reduced time refining the draft and surfaced structural issues earlier than solo writing would have. The user presented, directed, verified, and refined; Claude suggested and flagged.

Table 5: Generative AI Tool Usage — Grammarly

Field	Description
1. GenAI tool used	Grammarly.
2. Purpose of GenAI tool for this assignment	Grammar correction and sentence structure checking on paragraphs after initial drafting.
3. Prompts or instructions given	No explicit prompts. Grammarly operates by analysing pasted or typed text and surfacing inline suggestions. Text was submitted by pasting draft paragraphs into the Grammarly editor.
4. Amount of prompts or instructions given	Not applicable (non-prompt-based tool). Used across a small number of editing passes on selected paragraphs.
5. Summary of GenAI output	Grammarly flagged potential grammatical errors, suggested wording alternatives, and highlighted sentences flagged as unclear or overly long.
6. GenAI output verification	All suggestions were reviewed individually. Grammar corrections were accepted only where they were grammatically correct and preserved the intended meaning. Some suggestions (particularly structural rewrites) were rejected where they altered the meaning or register of the original sentence.
7. GenAI output usage or modification	Accepted grammar and punctuation corrections were applied directly. Sentence-structure suggestions were used as prompts to reconsider phrasing, but the user typically wrote their own reformulations rather than accepting Grammarly’s suggested rewrites verbatim.
8. Reflection on generated content and usage	What worked well: Grammarly was useful as a secondary check for typos and minor grammatical issues that can be easy to miss in self-editing. What did not work well: Suggested rewrites sometimes reduced specificity or altered technical meaning, particularly in sentences containing domain-specific terminology. Value added: Grammarly acted as a final pass for surface-level correctness, complementing the substantive drafting and editing done with Claude.